COMPLETE GUIDE · ENGLISH

Rovunet Panel

Your own multi-protocol access server, in one install

Installation, protocols, users, limits, subscription links, operations and troubleshooting: everything needed to run Rovunet Panel on your own Linux server.

14	443	1	5 min
protocols	fronted port	link per user	limit checks

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License **AGPL-3.0**

Community **t.me/RovunetVPN**

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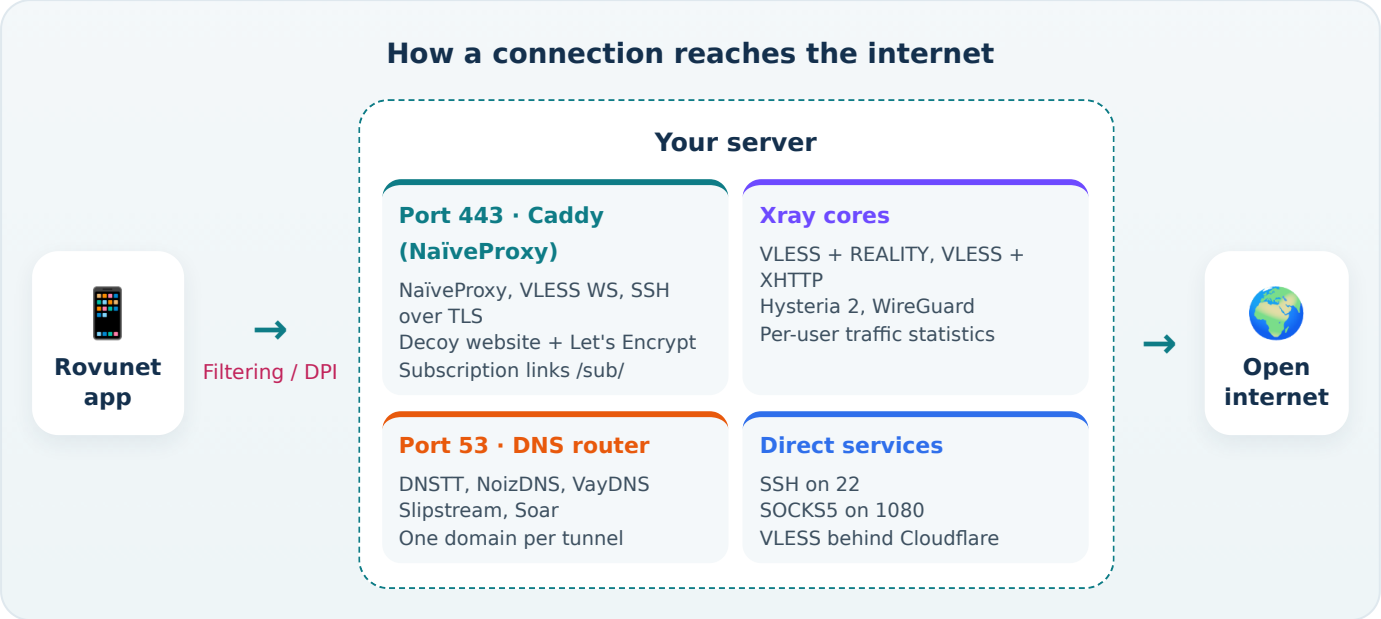
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01 Overview

Rovunet Panel is an open-source command-line tool that turns a Linux server into a multi-protocol access server. It installs every protocol as a systemd service, creates users and prints ready-to-import configurations, so each user has several independent ways in instead of depending on one protocol or one network condition.



Key features

14 protocols, one tool

Proxies, SSH, DNS tunnels and Xray cores, installed and managed from one command line with an interactive menu.

One login per user

The same username and password work on every tunnel; VLESS identities are per user and revoked within seconds.

Subscription link

Every user gets one HTTPS link with all of their configurations for the app's Subscriptions menu.

Time and traffic limits

Unlimited or limited users by days and/or GB, enforced every 5 minutes, renewable with the same link.

Hidden on port 443

VLESS and SSH over TLS share NaïveProxy's domain, port and real certificate behind a decoy website.

Several addresses

Extra IPs of the same server are added to every link, so one blocked IP does not cut users off.

Safe operations

Checksum-verified updates with rollback and full backup and restore, including used traffic.

Diagnostics

Live dashboard, health checks, certificate expiry, core statistics and per-service logs.

02 Protocols

Each protocol travels differently, so different networks let different ones through. Offer users several, ordered from the fastest to the most resilient.

VLESS WS · VLESS + REALITY · VLESS + XHTTP · Hysteria 2 · WireGuard · NaïveProxy · SSH over TLS · SSH · SOCKS5 · DNSTT · NoizDNS · VayDNS · Slipstream · Soar

Protocol	Carried over	Port	Best for
VLESS over WebSocket (fronted)	WebSocket + TLS, real certificate	443/tcp	First choice where HTTPS works
VLESS over WebSocket (Cloudflare)	WebSocket through Cloudflare's CDN	2053/tcp	When the server's IP is blocked
NaïveProxy	HTTP/2 + TLS, decoy website	443/tcp	Where TLS to your domain works
SSH over TLS (StunTLS)	SSH inside TLS / WebSocket	443/tcp	An alternative on the same port
SSH	Plain SSH	22/tcp	Where port 22 is open
SOCKS5	Plain SOCKS5, unencrypted	1080/tcp	Open networks only
DNSTT / NoizDNS	Encrypted DNS tunnel	53/udp	Restricted networks where DNS passes
VayDNS	DNS tunnel, tunable wire format	53/udp	Like DNSTT, with more tuning
Slipstream	QUIC over DNS	53/udp	Experimental
Soar	Multipath DNS with resolver discovery	53/udp + tcp	Unevenly filtered resolvers
VLESS + REALITY	TCP, imitates a real TLS site	any TCP	Where TLS to well-known sites passes
VLESS + XHTTP	HTTP requests over TLS	any TCP	Behind a CDN or where WebSocket is blocked
Hysteria 2	QUIC	any UDP	Lossy networks where UDP passes
WireGuard	UDP	any UDP	Where WireGuard is not blocked

Which protocol should I offer first?

Does HTTPS to your own domain on port 443 work? — yes ▶ **VLESS WS · NaïveProxy · SSH over TLS**
Fastest and most dependable; all share port 443 behind the decoy site.

▼ no

Is only the server's IP blocked? — yes ▶ **VLESS behind Cloudflare**
Traffic goes through Cloudflare's CDN instead of the blocked IP.

▼ no

Does TLS to well-known sites still pass? — yes ▶ **VLESS + REALITY · VLESS + XHTTP**
REALITY borrows a popular site's identity; its target is picked automatically.

▼ no

Does UDP pass? — yes ▶ **Hysteria 2 · WireGuard**
Good on lossy links when UDP is not dropped.

▼ no

Only DNS works → **DNSTT · NoizDNS · VayDNS · Soar** (slow, but often the last path that works during a shutdown)

✓ **Order of configurations**

`users configs` and subscription links list configurations from the fastest to the slowest: VLESS WS, XHTTP, REALITY, NaïveProxy, SSH over TLS, SSH, then UDP-based and unencrypted ones, and DNS tunnels last. The app's own HTTPS test then ranks them on the user's network.

03 Before you install

Two things decide most installations: a server whose provider does not block the ports you need, and DNS records created before installing.

Item	Requirement
Operating system	Linux with systemd: Ubuntu 20.04+ or Debian 10+
Access	Root (or sudo) over SSH
Domain	A domain whose DNS records you control (needed for DNS tunnels, NaïveProxy, VLESS and TLS-based cores)
Hardware	1 vCPU, 1 GB RAM, 20 GB disk for a small group; bandwidth matters more than CPU
Provider	Inbound UDP/53 allowed for DNS tunnels; no port restrictions; generous bandwidth

Ports

Port	Used by
53/udp, 53/tcp	DNS tunnels (DNSTT, NoizDNS, VayDNS, Slipstream, Soar)
80/tcp, 443/tcp	NaïveProxy and everything fronted by it (VLESS, SSH over TLS, subscription links)
22/tcp	SSH
1080/tcp	SOCKS5 (open networks only)
2053/tcp (or 2083, 2087, 2096, 8443)	VLESS behind Cloudflare
any free TCP/UDP port	Xray cores (REALITY 9443, XHTTP 2096, Hysteria 2 8443/udp, WireGuard 51820/udp by default)

DNS records

Create the records for the protocols you plan to use **before** installing. Every record is **DNS only** (grey cloud in Cloudflare): Caddy must be reached directly for its certificate, and DNS tunnels need raw DNS at the server.

Type	Name	Points to
A	`ns.example.com`	`SERVER_IP`
NS	`t.example.com`	`ns.example.com`

Type	Name	Points to
NS	ts.example.com	ns.example.com
NS	v.example.com / vs.example.com	ns.example.com
NS	s.example.com / ss.example.com	ns.example.com
NS	r.example.com / rs.example.com	ns.example.com
A	example.com	SERVER_IP

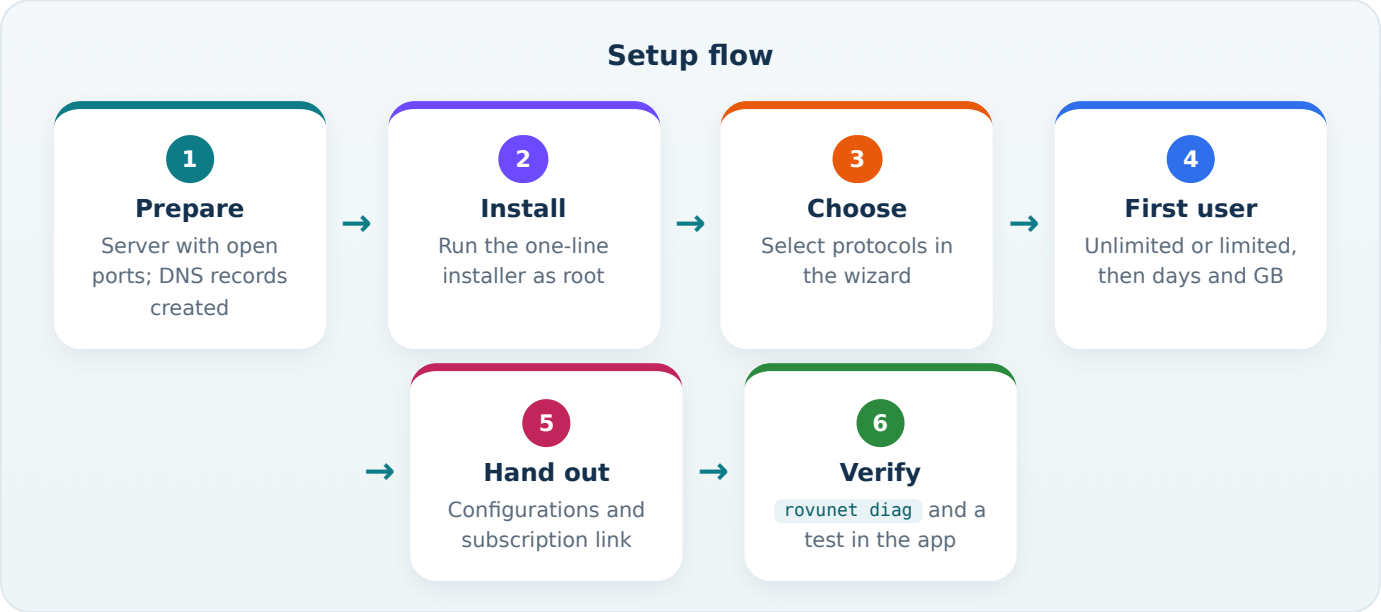
- `ns.example.com` is the name server that answers all DNS tunnels.
- `t`, `ts`, `v`, `vs`, `s`, `ss`, `r`, `rs` are delegated subdomains: DNSTT/NoizDNS, VayDNS, Slipstream and Soar, each with a SOCKS5 (or direct) and an SSH backend.
- `example.com` itself serves NaïveProxy, the fronted VLESS and SSH over TLS, and subscription links.
- Minimal setups: NaïveProxy only needs the `A example.com` record; DNSTT only needs `A ns` and `NS t`.

Check propagation (minutes to a few hours)

```
dig +short A ns.example.com
dig +short NS t.example.com
```

04 Installation and first setup

The installer is interactive: it asks which protocols to enable and does the rest.



Online install (recommended)

```
curl -fsSL https://raw.githubusercontent.com/msVPNco/rovunet-panel/main/install.sh | sudo bash
```

1. **Run the installer**

   It downloads the binaries for the selected protocols, generates keys, creates systemd services and firewall rules.

2. **Choose protocols**

   NaïveProxy plus VLESS is the recommended base: VLESS and SSH over TLS are fronted on port 443 from the start. REALITY and XHTTP are recommended among the Xray cores.

3. **Create the first user**

   Enter a username and password, then answer **Unlimited or Limited**. For a limited user enter days and GB; each defaults to unlimited.

4. **Copy the output**

   The installer prints every configuration of the first user and the user's subscription link. Send them over a private channel.

5. **Check the server**

   Run `sudo rovunet diag`. Every tunnel should be active, every port listening and every key present.

Everyday commands

```
sudo rovunet                # interactive menu
sudo rovunet users add      # add a user (unlimited or limited)
sudo rovunet users configs  # every configuration + subscription link
sudo rovunet diag           # check services, ports, keys, DNS
```

Running the installer again

On a configured server the installer keeps the existing tunnels, adds only new protocols and prints all configurations again. To add a second tunnel of a type you already have, use `sudo rovunet tunnel add`.

Xray cores

REALITY, XHTTP, Hysteria 2 and WireGuard are Xray cores. The installer offers them next to the other protocols; they can also be added later. XHTTP and Hysteria 2 use the certificate Caddy obtained for the NaïveProxy domain, so they need NaïveProxy too. REALITY chooses its camouflage target automatically from a bank of high-traffic sites that are rarely blocked.

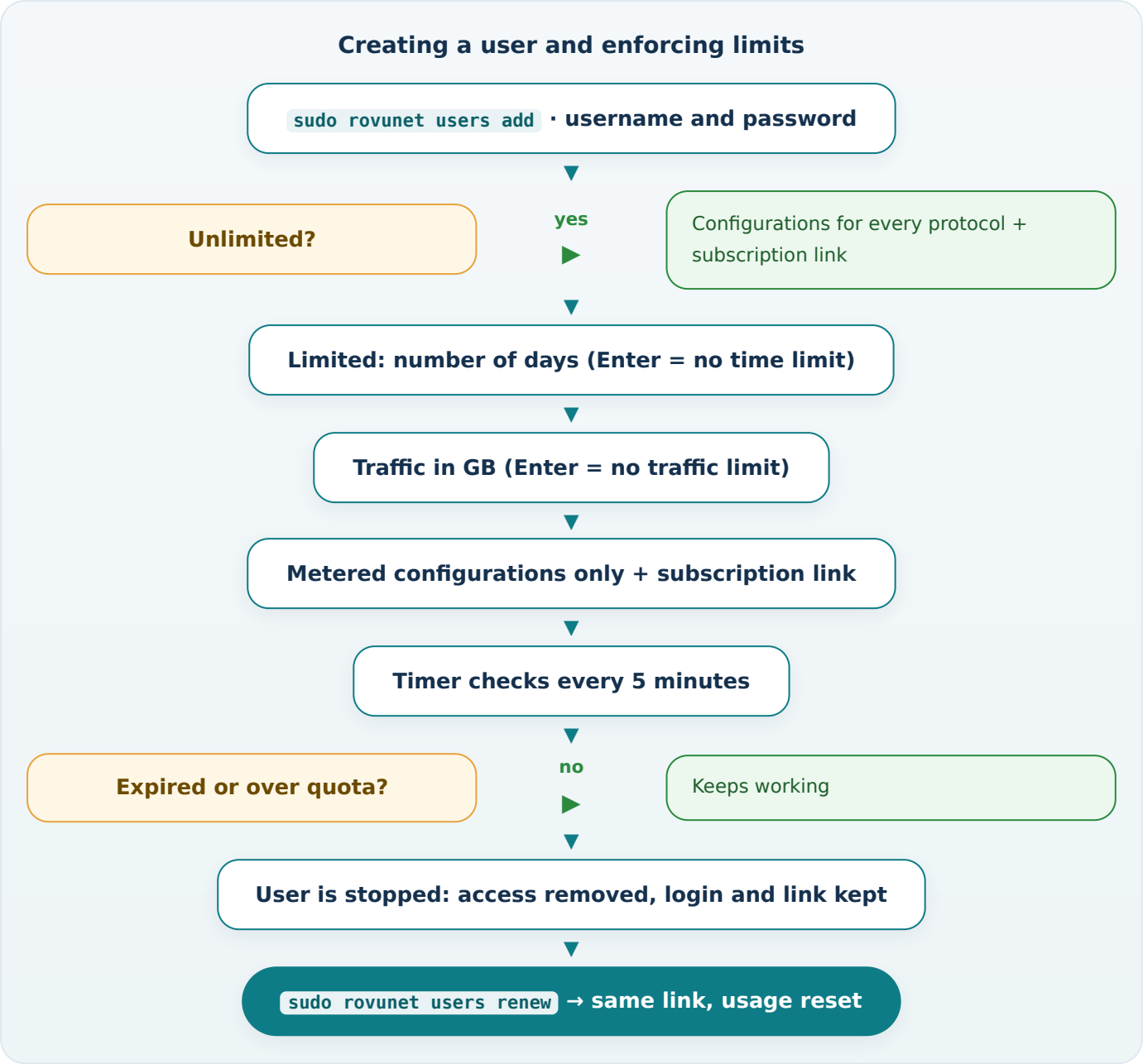
```
sudo rovunet core-install      # add REALITY, XHTTP, Hysteria 2 or WireGuard
sudo rovunet core-list         # name, protocol, port
sudo rovunet core-remove r1    # stop and delete a core
```

Offline install

If the server cannot reach GitHub, download the panel and the transport binaries on another computer, copy them to `/tmp/rovunet/` and run `sudo rovunet install --bin-dir /tmp/rovunet`. Use `arm64` builds on ARM servers.

05 Users, limits and subscription links

Users are global: one username and password work on every tunnel. The protocol belongs to the tunnel, not the user.



Managing users

```
sudo rovunet users add                              # one user
sudo rovunet users bulk_add --count=50 --prefix=user  # many users
sudo rovunet users list                             # limits, usage, stopped users
sudo rovunet users renew                            # bring a stopped user back
sudo rovunet users remove                           # remove a user
```

What is counted

A traffic limit can only apply to traffic the panel measures, so a limited user receives only the counted configurations. 1 GB is 1024^3 bytes; upload and download both count.

Counted · given to limited users	Not counted · unlimited users only
VLESS over WebSocket	SSH, SSH over TLS, NaïveProxy
VLESS + REALITY and VLESS + XHTTP (while the core's statistics answer; `diag` checks this)	DNS tunnels, Soar, Hysteria 2, WireGuard

Subscription links

Every user, limited or not, has one link with all of their configurations: `https://example.com/sub/<token>.json`. It is printed after the user is created and by `sudo rovunet users configs`. Paste it into the app's **Subscriptions** menu.

- Served as a static JSON file by the NaïveProxy site on port 443; it needs a NaïveProxy tunnel and nothing else.
- Kept up to date automatically after any user or tunnel change, after updates and every 5 minutes for limited users.
- The token is random; treat the link as a secret, like the configurations themselves. A password change issues a new link.
- A stopped user's link delivers no new configurations until `users renew`.

Several addresses for one server

If one IP of the server is blocked, other IPs of the same server may still work. List them after the main address; subscription links then carry every address-based configuration once per extra address.

```
sudo rovunet config address auto,198.51.100.9   # main address + extra IP
sudo rovunet config address auto                # remove the extras
```

Each extra address must reach this server

Use an IP configured on this server's network interface (or a hostname pointing to it) with the same ports open. This adds addresses, not locations.

Security

Never hand out `root` or any account with a shell as a VPN login. Give every person their own user, so one lost phone can be revoked alone, and send configurations only over private channels.

06 Operations

Day-to-day work is a handful of commands. All of them are safe to repeat.

Maintenance

```
sudo rovunet update                          # verified update with rollback
sudo rovunet backup /root/rovunet-backup.txt  # full backup (keys, users, usage)
sudo rovunet restore /root/rovunet-backup.txt # restore on this or a new server
sudo rovunet stats                           # live dashboard
sudo rovunet tunnel status                   # every tunnel and core
sudo rovunet tunnel logs <tag>               # one service log
```

Updating

`sudo rovunet update` downloads the latest release, verifies it against the release's SHA256SUMS, replaces the binary atomically and rolls back if installation fails. Tunnels, users and keys are untouched and existing links keep working.

Backup and restore

A backup holds everything needed to run the setup elsewhere: configuration, every tunnel's keys and certificates, all users and passwords, the Xray cores and each user's traffic used so far. It is one block of text; keep it private.



Server to server in one line

```
ssh root@old rovunet backup - | ssh root@new rovunet restore - --yes
```

Monitoring

Command	Shows
`sudo rovunet stats`	Live dashboard: CPU, RAM, traffic, connections per protocol

Command	Shows
`sudo rovunet tunnel status`	Every tunnel and Xray core with its service state; certificate expiry
`sudo rovunet diag`	Services, ports, keys, DNS, boot persistence, core statistics
`sudo rovunet tunnel logs <tag>`	A tunnel's service log

Command reference

Command	Purpose
`sudo rovunet`	Interactive menu
`rovunet install`	Install or add protocols
`rovunet update`	Verified update with rollback
`rovunet users add | bulk_add | list | renew | remove | configs`	User management
`rovunet tunnel add | remove | start | stop | share | status | logs | edit`	Tunnels
`rovunet core-install | core-list | core-remove`	Xray cores
`rovunet config address | export | import`	Addresses and configuration
`rovunet router status | mode | switch`	DNS router
`rovunet backup | restore`	Backup and restore
`rovunet diag | stats | mtu | restart`	Health and tuning
`rovunet expire-users`	Apply limits now (runs every 5 minutes)

07 Troubleshooting

Start with `sudo rovunet diag`: it checks services, ports, keys, DNS and boot persistence and names the failing part.

Server

Symptom	Likely cause	Fix
Tunnel not `active`	Port, key or configuration problem	`rovunet diag`, then `rovunet tunnel logs <tag>`
Port 53 already in use	systemd-resolved stub listener	Disable it and set `nameserver 1.1.1.1` in `/etc/resolv.conf`
`dig` to the server times out	Provider blocks inbound UDP/53	Change provider or use port-443 protocols
Let's Encrypt fails	A record not propagated or port 80 closed	Wait, open port 80, then `rovunet restart`
Only one DNS tunnel works	Single routing mode	`rovunet router status`, switch with `router mode`

Client

Symptom	Likely cause	Fix
Connects but no traffic	Network path or client issue	Test from the server with `curl -x socks5h://…`; if it works there, the server is fine
Connection was reset	Filtering on that port	Front the tunnel with NaïveProxy on 443
VLESS connects, no site opens	User removed or password changed	Hand out a fresh configuration or subscription link
Limited user lost REALITY/XHTTP	Core statistics not answering	`rovunet diag`; the feed refreshes within 5 minutes
Slow DNS tunnel	MTU too large for the path	`rovunet mtu 1200` (1100 for very poor links)
Cloudflare 521/522 · 525/526	Port closed · wrong SSL mode	Open the port · set SSL/TLS to Full

Getting help
Ask in t.me/RovunetVPN with the panel version, the protocol and the `diag` output. Remove passwords, keys and full links before posting anything publicly.

08 Resources and support

Telegram

t.me/RovunetVPN · news, questions and help

Panel source

github.com/msVPNco/rovunet-panel

Android app

github.com/msVPNco/rovunet-android

Documentation

Full docs in English, Persian, Russian and Chinese in the repository's docs folder

Rovunet is unpaid volunteer work. Voluntary donations: **USDT on Tron (TRC20) only** to `TG3Mhou3pdq9KaFgJSHjDiDx2ahhJuHvbJ`.

License: AGPL-3.0. Rovunet Panel is based on SlipGate by Anonvector and is not affiliated with it; see NOTICE for attributions.

Early release, no guarantees

Rovunet Panel is under active development. Test the protocols you need on your own network before relying on them and keep a backup. Which protocols work depends on the server, the operator and the network; nothing here guarantees access through every network.